

# Midterm 1

Math 124, Fall 2026

Name: \_\_\_\_\_

username: \_\_\_\_\_

UMID: \_\_\_\_\_

Room: ☐ 1360 East Hall    ☐ 3088 East Hall

## Instructions

- This exam consists of 8 problems across 12 pages, worth a total of 100 points.
- You have 120 minutes to complete this exam, unless you have extended-time accommodations through SSD.
- Write your username in the top right corner of each page where a space is provided.
- For free response problems, **you must show all of your work**, and write your final answer in the small box when one is provided.
- For multiple choice problems, completely fill in bubbles and square boxes.
  - ☐ A bubble means that you should only select one choice.
  - ☐ A square box means that you should select all that apply.
- You may refer to **one double-sided 4 × 6-inch index card**, handwritten directly on the card. Other than that, you may not refer to any other resources or technology during the exam (no phones, watches, or calculators).

By signing below, I affirm that this work is my own and that I have followed the exam's resource and collaboration rules.

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**Problem 1 (10 pts)**

Let

$$\vec{d} = \begin{bmatrix} 6 \\ -9 \\ 18 \end{bmatrix}$$

You may leave numbers as unsimplified fractions.

- a) (4 pts) Find a unit vector pointing in the same direction as  $\vec{d}$ .

unit vector =

- b) (3 pts) Find a vector of length 6 pointing in the direction opposite to  $\vec{d}$ .

vector =

- c) (3 pts) A robot starts at  $P = (5, -6, 8)$  and moves  $x$  units in the direction of  $\vec{d}$ , ending at  $(9, -12, 20)$ . What is the value of  $x$ ?

$x =$

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**Problem 2 (14 pts)**

Suppose  $\vec{u}, \vec{v} \in \mathbb{R}^3$  satisfy

$$\|\vec{u}\| = 3, \quad \|\vec{v}\| = 4, \quad \vec{u} \cdot \vec{v} = -10$$

- a) (6 pts) Find the value of  $c$  for which  $\vec{u} + c\vec{v}$  is orthogonal to  $2\vec{u} - \vec{v}$ .

$c =$

- b) (5 pts) Find  $\|3\vec{u} + \vec{v}\|$ .

$\|3\vec{u} + \vec{v}\| =$

- c) (3 pts) Find the angle between  $\vec{u}$  and  $\vec{v}$ . Then select the angle type.

$\text{angle} = \cos^{-1} \left( \div style="border: 1px solid black; width: 100px; height: 70px; display: flex; align-items: center; justify-content: center;">$

☐ Acute   ☐ Right   ☐ Obtuse

**Problem 3 (15 pts)**

Consider the line  $\ell$  in  $\mathbb{R}^2$  given by

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \end{bmatrix} + t \begin{bmatrix} 6 \\ 5 \end{bmatrix}, \quad t \in \mathbb{R}$$

- a) (4 pts) Write an equation for  $\ell$  in the form  $ax + by = c$ .

equation:

- b) (7 pts) Which of the following are valid scalar-parametric forms of  $\ell$ ? **Select all that apply.** In each option,  $t \in \mathbb{R}$ .

☐  $\begin{cases} x = 5 + 12t \\ y = 4 + 10t \end{cases}$

☐  $\begin{cases} x = 11 + 12t \\ y = 8 + 10t \end{cases}$

☐  $\begin{cases} x = -1 - 18t \\ y = -2 - 15t \end{cases}$

☐  $\begin{cases} x = 11 + 5t \\ y = 8 - 6t \end{cases}$

- c) (4 pts) Let  $\ell'$  be the line perpendicular to  $\ell$  that passes through  $(7, -6)$ . Write an equation for  $\ell'$  in the form  $ax + by = c$ .

equation:

**Problem 4 (11 pts)**

Let  $\vec{u}_1$  and  $\vec{u}_2$  form an orthonormal basis of  $\mathbb{R}^2$ . Suppose

$$\vec{w} = 6\vec{u}_1 + 2\vec{u}_2.$$

a) (4 pts) Find  $\|\vec{w}\|$ .

$\|\vec{w}\| =$

b) (3 pts) Find  $\vec{w} \cdot \vec{u}_1$ .

$\vec{w} \cdot \vec{u}_1 =$

c) (4 pts) Suppose  $\vec{v}$  is another vector in  $\mathbb{R}^2$ . Given that the angle between  $\vec{v}$  and  $\vec{u}_1$  is  $35^\circ$ , which of the following *could be* the angle between  $\vec{v}$  and  $\vec{u}_2$ ? **Select all that apply.**

- ☐  $35^\circ$   
 ☐  $55^\circ$   
 ☐  $90^\circ$   
 ☐  $125^\circ$   
 ☐  $145^\circ$

**Problem 5 (12 pts)**

Consider the vectors

$$\vec{u}_1 = \frac{1}{9} \begin{bmatrix} 4 \\ 4 \\ 7 \end{bmatrix}, \quad \vec{u}_2 = \frac{1}{9} \begin{bmatrix} -8 \\ 1 \\ 4 \end{bmatrix}, \quad \vec{w} = \begin{bmatrix} 4 \\ 2 \\ 3 \end{bmatrix}$$

The vector  $\vec{w}$  is in  $\text{span}(\vec{u}_1, \vec{u}_2)$ .

- a)** (6 pts) Verify that  $\vec{u}_1$  and  $\vec{u}_2$  are orthogonal and each has length 1.

- b)** (6 pts) Write  $\vec{w}$  as a linear combination of  $\vec{u}_1$  and  $\vec{u}_2$ .

$\vec{w} =$  $\vec{u}_1 +$  $\vec{u}_2$

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**Problem 6 (16 pts)**

Let  $P$  be the plane through  $(6, -5, 8)$  with direction vectors

$$\vec{u} = \begin{bmatrix} 2 \\ 5 \\ 3 \end{bmatrix}, \quad \vec{v} = \begin{bmatrix} 4 \\ 5 \\ 1 \end{bmatrix}$$

- a) (8 pts) Write an equation for  $P$  in the form  $ax + by + cz = d$ .

equation:



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Recall that  $P$  is the plane through  $(6, -5, 8)$  with direction vectors

$$\vec{u} = \begin{bmatrix} 2 \\ 5 \\ 3 \end{bmatrix}, \quad \vec{v} = \begin{bmatrix} 4 \\ 5 \\ 1 \end{bmatrix}$$

b) (8 pts) For each vector-parametric equation below, select whether it is a valid description of  $P$  and give a brief justification. In every option,  $s, t \in \mathbb{R}$ .

*Hint: Use the equation you found in the previous part.*

i) (2 pts)  $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ -5 \\ 8 \end{bmatrix} + s(\vec{u} + \vec{v}) + t\vec{v}$

☐ Valid ☐ Invalid

ii) (2 pts)  $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 9 \\ -4 \\ 6 \end{bmatrix} + s\vec{u} + t\vec{v}$

☐ Valid ☐ Invalid

iii) (2 pts)  $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ -5 \\ 9 \end{bmatrix} + s\vec{u} + t\vec{v}$

☐ Valid ☐ Invalid

iv) (2 pts)  $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ -5 \\ 8 \end{bmatrix} + s(2\vec{u} - 3\vec{v}) + t(4\vec{u} - 6\vec{v})$

☐ Valid ☐ Invalid

**Problem 7 (10 pts)**

The equations

$$P_1 : 5x + 3y = 10, \quad P_2 : 2x + 6y + 4z = 20$$

describe two planes in  $\mathbb{R}^3$ . Write the set of common solutions to these two equations in vector-parametric form.

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} =$$

**Problem 8 (12 pts)**

Let

$$\vec{v} = \begin{bmatrix} 7 \\ 5 \\ 6 \end{bmatrix}, \quad \ell = \text{span} \left( \begin{bmatrix} 4 \\ -5 \\ 7 \end{bmatrix} \right), \quad P : 4x - 5y + 7z = 0$$

- a) (6 pts) Find the orthogonal projection of  $\vec{v}$  onto  $\ell$ .

$\text{proj}_{\ell}(\vec{v}) =$

- b) (6 pts) Find the orthogonal projection of  $\vec{v}$  onto  $P$ .

$\text{proj}_P(\vec{v}) =$

**username:** \_\_\_\_\_

Congrats on finishing Midterm 1!

Feel free to draw us a picture about Math 124 in the box below (or use it for scratch work).

A large, empty rectangular box with a thin black border, intended for a drawing or scratch work related to Math 124.